

Java Exception Handling Report

NegativeArraySizeException

1. What is NegativeArraySizeException?

NegativeArraySizeException is a Java runtime exception that occurs when a program tries to create an array with a negative size. For example, `new int[-5]` is invalid because an array can have zero or a positive number of elements, but it cannot have a negative number of elements.

2. Objective

To understand and demonstrate how Java handles an attempt to create an array using a negative size, and to handle the `NegativeArraySizeException` using try-catch.

3. Task Description

Write a Java program that attempts to create an integer array with size -5. Since the array size is negative, Java throws `NegativeArraySizeException`. The exception must be caught and a suitable message should be displayed.

4. Step-wise Execution

1. Start the program.
2. Enter the `main()` method.
3. Set the array size variable to -5.
4. Try to create an integer array using `new int[size]`.
5. Java detects that the requested array size is negative and throws `NegativeArraySizeException`.
6. Control moves to the catch block.
7. The program prints: "Exception: Array size cannot be negative".
8. The program ends normally after handling the exception.

5. Algorithm

1. START
2. Set `size = -5`.
3. Begin try block.
4. Create an integer array of size `size`.
5. If the size is negative, catch `NegativeArraySizeException`.
6. Display the exception message.
7. STOP

6. UML Diagram



+ main(args : String[]) : void	
main()	Creates array with negative size
	Handles NegativeArraySizeException

UML interpretation: NegativeArrayDemo contains the main() method. The method attempts array creation and handles NegativeArraySizeException.

7. Program Code

```
class NegativeArrayDemo {
    public static void main(String[] args) {
        try {
            int size = -5;
            int[] numbers = new int[size];
            System.out.println(numbers.length);
        } catch (NegativeArraySizeException e) {
            System.out.println("Exception: Array size cannot be negative");
        }
    }
}
```

8. Code Explanation

- **class NegativeArrayDemo** – Defines the Java class.
- **public static void main(String[] args)** – The starting point of program execution.
- **try** – Contains code that may generate an exception.
- **int size = -5;** – Stores a negative array size.
- **int[] numbers = new int[size];** – Attempts to create an integer array. Because size is -5, Java throws NegativeArraySizeException.
- **System.out.println(numbers.length);** – This statement is not reached because the exception occurs on the previous line.
- **catch (NegativeArraySizeException e)** – Catches the specific exception.
- **System.out.println(...)** – Displays a clear message explaining the problem.

9. Output Screenshot

Program Output
Exception: Array size cannot be negative

Expected console output:

Exception: Array size cannot be negative